

ai-powered-programming --tools --agents

past | present | future

2026-02-17

Spokane, Washington

>preface

- This presentation is for...
 - curious people
 - veteran software developers
 - junior programmers
 - idea guys
 - total newbs
 - entrepreneurs
 - vibe coders

>preface --not

- This presentation is NOT...
 - totally comprehensive
 - exploring Chinese AI products
 - addressing ethical concerns about AI training
 - addressing employment concerns about AI replacement
 - addressing AI doom concerns about existential risk

>questions

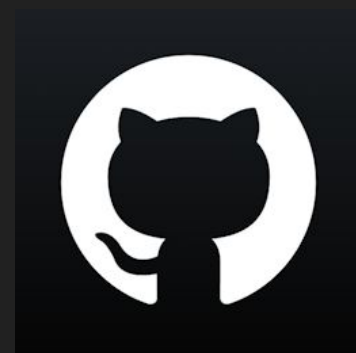
- Shorter questions?
 - Please raise your hand.
- Longer questions?
 - Please wait until the end.

>outline

- Lineage of modern LLM technology
- Recent boom of AI products and services
- Current landscape of AI tools and agents
- Demos for the most popular products
- Best practices and recommendations
- Links and bookmarks
- What's coming next?

>whoami

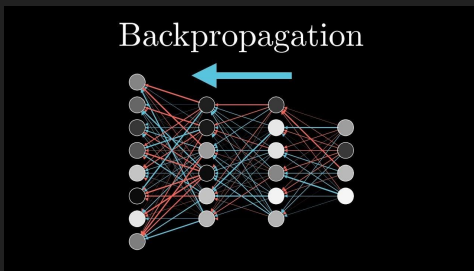
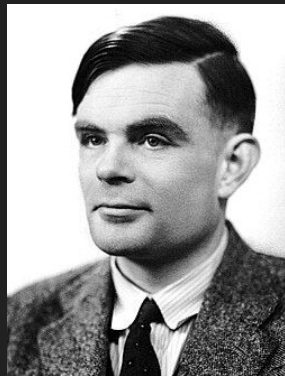
- John W. Shaffstall
 - 15 years of professional software dev experience
 - Native mobile apps, websites, APIs
 - Game content pipelines, 3D rendering
 - Vibe coding
 - Links
 - <https://x.com/JohnShaffstall>
 - <https://github.com/jwshaffstall>
 - Discord
 - John S. in the Spokane Tech Discord
 - <https://discord.gg/DVkaXmKCTj>



```
>history
```

>lineage --tech llm

- 1943: Artificial neural networks
- 1950: Turing Test thought experiment
- 1957: Perceptron
- 1966: ELIZA chat program
- 1974-1980: AI winter
- 1986: Backpropagation
- 1987-2000: Another AI winter



```
Welcome to
EEEEEE LL IIII ZZZZZZ AAAAA
EE LL II ZZ AA AA
EEEE LL II ZZ AAAAAA
EE LL II ZZ AA AA
EEEEEE LLLLLL IIII ZZZZZZ AA AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU: Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU: They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU: Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU: He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU: It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

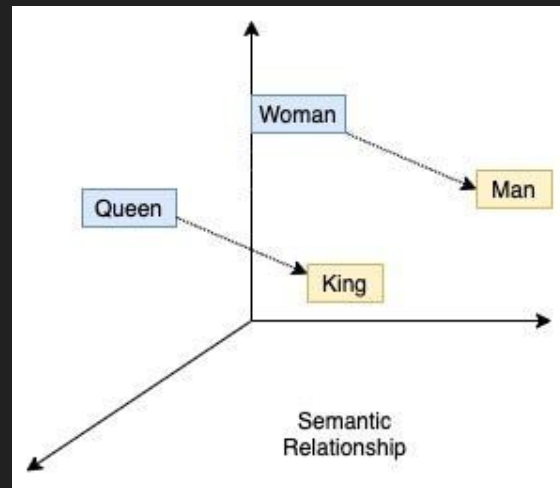
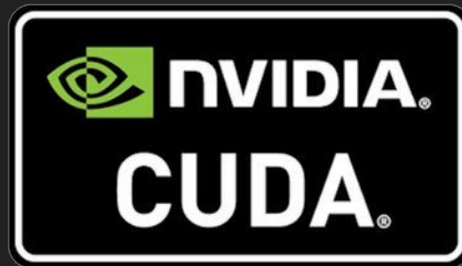

>lineage --tech gamers

- Graphics Processing Unit (GPU)
 - 80-90s: Silicon Graphics
 - 90s: 3dfx, **NVIDIA**, ATI
- 2001: Black & White (Lionhead)
 - Creature with basic AI learning
 - Demis Hassabis -> Deepmind (2010)
- Google Deepmind
 - 2015: AlphaGo
 - 2017: AlphaZero (self-play)
 - 2019: AlphaStar (StarCraft II)
 - **2018-2024: AlphaFold (Nobel Prize)**



>timeline --tech llm

- 2007: NVIDIA CUDA
 - General-purpose GPU compute
- 2011-2012: AlexNet image classification
- 2013: word2vec
- 2017: Google "Attention Is All You Need"
 - Transformers for machine translation
- 2018: Google BERT
 - Natural human language for search queries
- 2018: OpenAI GPT-1
- 2019: OpenAI GPT-2



```
>timeline --openai 2020-2022
```

- May 2020: OpenAI GPT-3
- January 2021: OpenAI DALL·E
- April 2022: OpenAI DALL·E 2
- September 2022: OpenAI Whisper
- **November 2022: OpenAI GPT-3.5 (ChatGPT)**



>timeline --tech ai-models 2023

- Feb 2023: Meta LLaMA
- March 2023: Anthropic Claude
- March 2023: OpenAI GPT-4
- March 2023: Google Bard
- July 2023: Anthropic Claude 2
- July 2023: Meta Llama 2
- September 2023: Mistral 7B
- November 2023: OpenAI GPT-4 Turbo
- December 2023: Google Gemini 1.0
- December 2023: Mistral 8x7B
- December 2023: xAI Grok-1

>timeline --tech ai-models 2024

- February 2024: Google Gemini 1.5
- March 2024: Anthropic Claude 3
- April 2024: Meta Llama 3
- May 2024: OpenAI GPT-4o (multi-modal)
- June 2024: Anthropic Claude 3.5 Sonnet
- August 2024: xAI Grok 2
- Sep 2024: OpenAI o1 (reasoning)

>timeline --tech ai-models 2025

- **January 2025: DeepSeek-R1**
- February 2025: Gemini 2.0, Grok 3, Claude 3.7 Sonnet
- March 2025: Google Gemini 2.5 Pro
- April 2025: OpenAI GPT-4.1, o3, o4-mini
- April 2025: Meta Llama 4
- May 2025: Anthropic Claude 4
- July 2025: xAI Grok 4
- August 2025: OpenAI GPT-5
- September 2025: Anthropic Claude 4.5 Sonnet
- November 2025: Google Gemini 3 Pro
- **November 2025 Anthropic Claude 4.5 Opus**

>timeline --tech ai-tools

- **June 2021: GitHub Copilot**
- September 2022: Replit Ghostwriter
- October 2022: Codeium
- **November 2022: ChatGPT**
- December 2022: Perplexity (search)
- February 2023: Bing Chat
- **March 2023: Cursor**
- June 2023: Aider
- August 2023: JetBrains AI Assistant
- March 2024: Devin “AI software engineer”
- April 2024: Gemini Code Assist
- July 2024: Cline

```
>ls ~/ai-agents --year 2024
```

- September 2024: Replit Agent (cloud)
- November 2024: Cursor Agent Mode (IDE)
- November 2024: Windsurf (IDE agent)
- December 2024: Google Jules (cloud agent)


```
>ls ~/ai-agents --year 2025
```

- January 2025: JetBrains Junie (IDE Agent)
- **February 2025: Claude Code (CLI agent)**
- March 2025: Manus (cloud agent)
- April 2025: OpenAI Codex (CLI agent)
- May 2025: Amp, OpenCode (CLI agents)
- May 2025: GitHub Copilot Agent (cloud)
- June 2025: Gemini CLI
- November 2025: Clawdbot / Moltbot / OpenClaw (personal agent)

>demos

>setup --recommend tools

- Git
 - <https://git-scm.com/>
- Node.js 22+ (with npm 11.x)
 - nvm (node version manager)
 - <https://github.com/nvm-sh/nvm>
- Python 3.x
 - <https://www.python.org/downloads/>
- Windows: PowerShell 7.x
 - <https://github.com/PowerShell/PowerShell>



>setup --recommend apps

- Visual Studio Code
 - <https://code.visualstudio.com/>
- GitHub Desktop app
 - <https://github.com/apps/desktop>
- JetBrains Toolbox
 - <https://www.jetbrains.com/toolbox-app/>
 - PyCharm
 - <https://www.jetbrains.com/pycharm/>
- Windows Terminal
 - <https://github.com/microsoft/terminal>



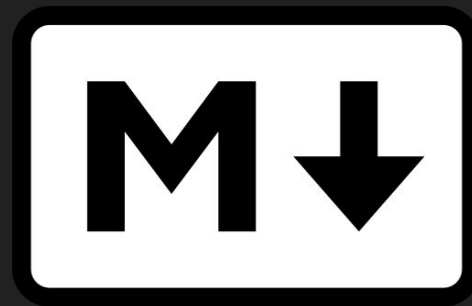
>demo @github/copilot

- GitHub Copilot IDE plugins (VS Code, JetBrains)
 - <https://github.com/features/copilot/ai-code-editor>
- GitHub Copilot CLI
 - <https://github.com/features/copilot/cli>
- GitHub Copilot Agents (Web)
 - <https://github.com/features/copilot/agents>
- GitHub Copilot Spark
 - <https://github.com/features/spark>
- Free and Paid tiers
 - <https://github.com/features/copilot/plans>



>techniques --basics

- Model selection
 - <https://artificialanalysis.ai/>
 - <https://arena.ai/leaderboard>
- Onboarding with markdown files
 - /init
 - AGENTS.md
 - CLAUDE.md
 - GEMINI.md
 - .github/copilot-instructions.md
- Tool modes
 - Ask / Chat
 - Agent
 - Plan
 - Debug



>demo @anysphere/cursor

- Cursor IDE (VS Code fork)
 - <https://cursor.com/>
- Cursor CLI
 - <https://cursor.com/cli>
- Free and Paid tiers
 - <https://cursor.com/pricing>



>demo @anthropic/claude



- Claude app (web and mobile)
 - <https://claude.ai/>
- Claude Code CLI
 - <https://code.claude.com/docs/en/overview>
- Claude Cowork (desktop app)
 - <https://claude.com/download>
- Free and Paid tiers
 - <https://claude.com/pricing>

The screenshot shows a terminal window titled "Claude Code" with a command prompt at "PS F:\dev\github\card-query-web> claude". The interface displays a welcome message "Welcome back John!" with a small robot icon and the text "Opus 4.6 · Claude Pro" and "F:\dev\github\card-query-web". To the right, there are sections for "Tips for getting started" (Run /init to create a CLAUDE.md file with instructions for Claude) and "Recent activity" (No recent activity). Below the welcome message, a list of commands is shown:

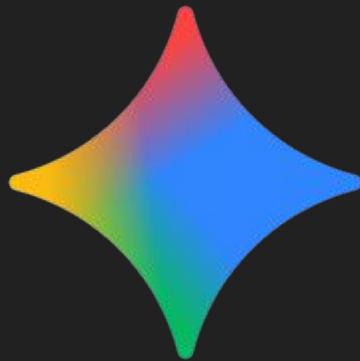
Command	Description
/install-slack-app	Install the Claude Slack app
/install-github-app	Set up Claude Github Actions for a repository
/init	Initialize a new CLAUDE.md file with codebase documentation
/insights	Generate a report analyzing your Claude Code sessions
/login	Sign in with your Anthropic account
/terminal-setup	Install Shift+Enter key binding for newlines

>demo @openai/codex

- Codex Agent (web and mobile)
 - <https://chatgpt.com/codex>
- Codex CLI
 - <https://developers.openai.com/codex/cli/>
- Codex App (Mac)
 - <https://openai.com/codex/>
- Free and Paid tiers
 - <https://developers.openai.com/codex/pricing/>



>demo @google/gemini

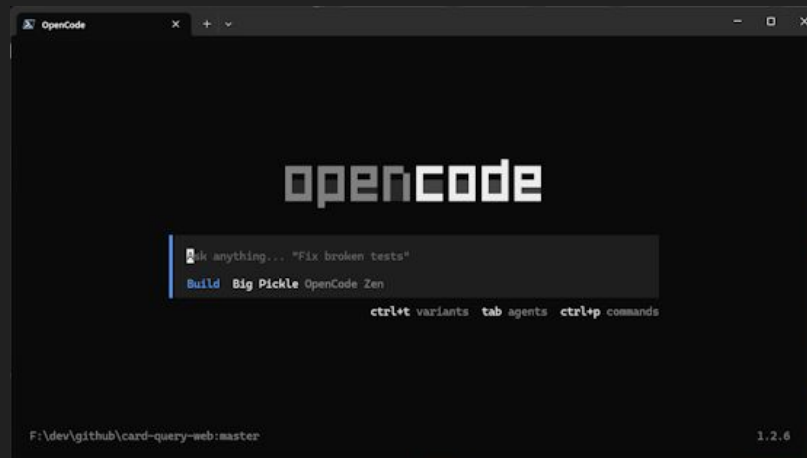


- Gemini Code Assist plugin (VS Code and JetBrains)
 - <https://codeassist.google/>
- Gemini CLI
 - <https://geminicli.com/>
- Google Jules (web)
 - <https://jules.google/>
- Google Antigravity (VS Code fork)
 - <https://antigravity.google/>
- Free and Paid tiers
 - <https://gemini.google/subscriptions/>

The screenshot displays the Gemini CLI interface in a terminal window. At the top, the word "GEMINI" is rendered in a large, stylized, multi-colored font. Below it, a list of tips for getting started is shown: 1. Ask questions, edit files, or run commands. 2. Be specific for the best results. 3. /help for more information. The interface then shows a prompt where the user has entered a command to write a short paragraph about why Gemini CLI is awesome. The response shows the AI starting to search the web for "Gemini CLI" to understand its main features and purpose. The bottom of the terminal shows the current directory as ~/code/gemini-cli (release*), the status as no sandbox (see /docs), and the model as gemini-2.5-pro (98% context left).

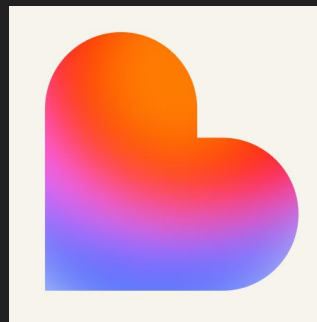
>demo @anomalyco/opencode

- OpenCode CLI
 - <https://opencode.ai/>
- Free and Paid tiers
 - <https://opencode.ai/zen>



>demo --web-tools

- Replit Agent
 - <https://replit.com/products/agent>
- Vercel v0
 - <https://v0.app/>
- Lovable
 - <https://lovable.dev/>



>demo @openclaw



- OpenClaw (CLI, Web, Chat)
 - <https://openclaw.ai/>
- @steipete was just hired by OpenAI
 - "OpenClaw will live in a foundation as an open source project that OpenAI will continue to support."



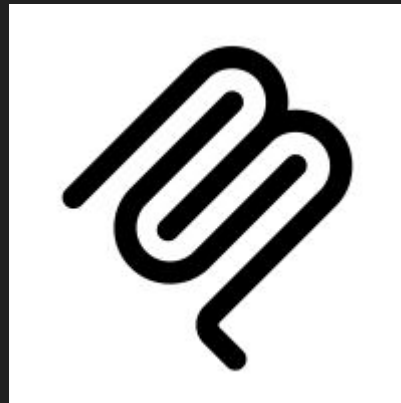
>demo --local-models

- LM Studio (desktop app, CLI)
 - <https://lmstudio.ai/>
 - <https://lmstudio.ai/docs/cli>
- ollama
 - <https://ollama.com/>
- llama.cpp
 - <https://github.com/ggml-org/llama.cpp>
- Hugging Face
 - <https://huggingface.co/>
- llm-checker (new!)
 - <https://github.com/Pavelevich/llm-checker>



>techniques --advanced

- Model Context Protocol (MCP)
 - <https://modelcontextprotocol.io/>
- Agent skills
 - <https://agentskills.io/home>
- Sub-agents
 - Claude Code
 - <https://code.claude.com/docs/en/sub-agents>
 - GitHub Copilot
 - <https://code.visualstudio.com/docs/copilot/agents/subagents>
- Ralph Loop
 - <https://github.com/snarktank/ralph>



>recommendations

>recommend --techniques

- use good models!
 - friends don't let friends use bad models
- /commands
 - /init for good onboarding
 - keep markdown files up-to-date
 - /plan for complex work
 - review the plan before proceeding
- manage scope and context
 - /new and specify targeted tasks for best results
- use scripts for automation
 - ask for scripts and tell it to use them
- document work
 - ask for docs and summaries in markdown files

>recommend --security

- don't be stupid
 - don't leak keys and access tokens
 - watch out for prompt injection
- sandbox
 - lowest necessary privileges
 - cloud agents when appropriate
- best practices
 - OWASP Top 10 for Agents
 - <https://genai.owasp.org/resource/owasp-top-10-for-agentic-applications-for-2026/>

>recommend --etiquette

- say please
 - have you said thank you once?
- disclose AI tool use
 - say which products and models were used
 - if the tool doesn't say, you can ask it
- pull requests / code reviews
 - don't flood open-source projects with AI slop
 - don't offload mental effort to other people

>recommend --free

- OpenCode CLI + free models
 - <https://opencode.ai/docs/zen/>
- GitHub Copilot + free tier (limited)
 - <https://github.com/features/copilot/plans>
- Cursor + free tier (limited)
 - <https://cursor.com/pricing>
- API: OpenRouter + free models
 - <https://openrouter.ai/collections/free-models>

>recommend --cheap

- ~\$8-10/month
 - **GitHub Copilot Pro**
 - ChatGPT Go + Codex
 - Google AI Plus
- ~\$20/month
 - **ChatGPT Plus + Codex**
 - **Cursor Pro**
 - **Google AI Pro**
 - **Claude Pro**

>recommend --expensive

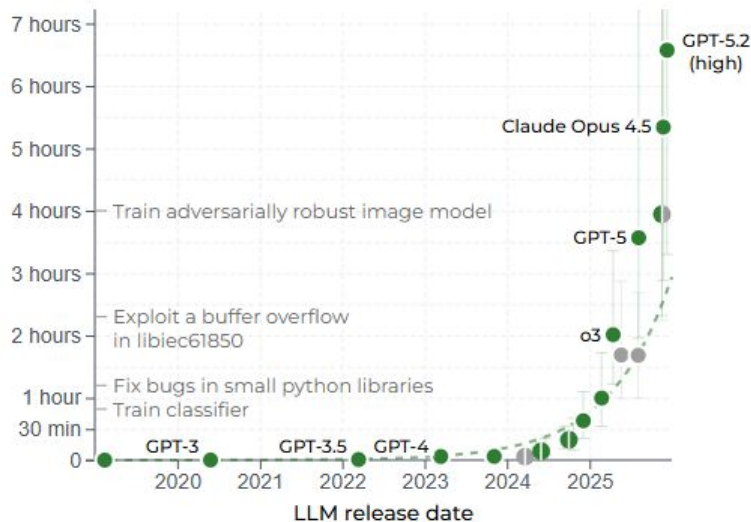
- ~\$30-60/month
 - GitHub Copilot Pro+
 - Cursor Pro+
 - X Premium+ / SuperGrok
- ~\$100/month
 - Claude Max 5x
- ~200+/month
 - Claude Max 20x
 - ChatGPT Pro + Codex
 - Google AI Ultra
 - SuperGrok Heavy

>future

>future

- Better AI models
- Longer context windows
- Better AI tools and capabilities
- Longer task completions
 - <https://metr.org/time-horizons/>
- Agent orchestration
- Agent swarms
- Autonomous corporations?

Time horizon of software tasks
different LLMs can complete
a 50% of the time



TH1.1 v

Log

Linear

50% Success

80% Success



>releases --year 2026

- February 5th
 - Anthropic Claude Opus 4.6
 - OpenAI GPT-5.3-Codex
- **February 17th (today!)**
 - xAI Grok 4.20 (4x agent swarm)
 - Anthropic Claude Sonnet 4.6
- Google
 - Gemini 3.x?
- Meta
 - Llama 5?

>bookmarks

>bookmarks --web

- Aggregators

- Hacker News

- <https://news.ycombinator.com/>

- Lobste.rs

- <https://lobste.rs/>

- Leaderboards

- Arena.ai (formerly LM Arena)

- <https://arena.ai/leaderboard>

- Artificial Analysis

- <https://artificialanalysis.ai/>

- Stats

- OpenRouter

- <https://openrouter.ai/>

- LLM Visualizations

- <https://bbycroft.net/llm>

- <https://writings.stephenwolfram.com/2023/02/what-is-chatgpt-doing-and-why-does-it-work/>

- Benchmarks

- ARC-AGI

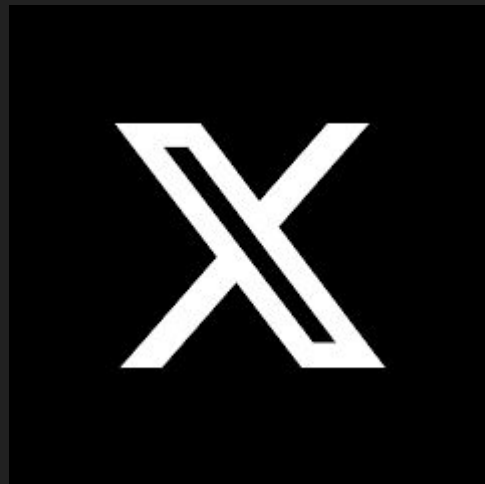
- <https://arcprize.org/arc-agi>

- Vibe Coding book (Rick Rubin)

- <https://www.thewayofcode.com/>

>bookmarks --x-formerly-twitter

- Andrej Karpathy: <https://x.com/karpathy>
- OpenAI:
 - <https://x.com/sama>
 - <https://x.com/gdb>
- Google
 - <https://x.com/demishassabis>
 - <https://x.com/OfficialLoganK>
- Claude Code: <https://x.com/bcherny>
- Three.js: <https://x.com/mrdoob>
- OpenClaw: <https://x.com/steipete>



>bookmarks --youtube

- AI Explained
 - <https://www.youtube.com/@aiexplained-official>
- Dwarkesh Patel
 - <https://www.youtube.com/@DwarkeshPatel>
- Lex Fridman
 - <https://www.youtube.com/lexfridman>
- Wes Roth
 - <https://www.youtube.com/@WesRoth>
- 3blue1brown
 - https://www.youtube.com/channel/UCYO_jab_esuFRV4b17AJtAw



>bookmarks --github

- OpenAI Codex CLI
 - <https://github.com/openai/codex>
- Google Gemini CLI
 - <https://github.com/google-gemini/gemini-cli>
- OpenCode CLI
 - <https://github.com/anomalyco/opencode>
- OpenClaw
 - <https://github.com/openclaw/openclaw>



>thanks

- Thank you to everyone
 - for showing up tonight!
- Thank you to David
 - for organizing this event!
- Thank you to the Spokane Go Users Group
 - for letting us share this event with you!
- Thank you to Limelyte
 - for letting us use this space!

>questions

✚ WOULD YOU LIKE TO KNOW MORE?

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